

Manuteja Medarametla

Data Engineer | Business Intelligence Engineer | Data Analyst
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SUMMARY

- Data Engineer with around **3 years of experience** building large-scale data systems across healthcare and finance domains, solving supply chain, risk, and reporting challenges by automating data pipelines, analytics workflows, and dashboarding at scale.
- Built and optimized distributed data pipelines in **PySpark** and **Databricks**, processing over 120M records weekly to drive stockout forecasting and risk analytics, cutting ETL runtime by 29% and improving planning accuracy by 22%.
- Designed robust data models using **SQL** (PostgreSQL, Synapse) and **dbt**, enabling clean joins across 15+ entities and reducing dashboard query time by over 80% through indexing, partitioning, and schema tuning, **Python (pandas, matplotlib)** for deep-dive analysis on expiry loss, cashflow patterns, and credit exposure — surfacing over \$4M in financial risk insights across two domains.
- Automated 50+ reports into dynamic **Power BI dashboards** with RLS and live refreshes, reducing manual work by 80% and delivering real-time insights for operations, finance, and compliance teams.
- Integrated structured and semi-structured feeds using **AWS Lambda, S3**, and Snowflake, improving FX data availability for financial risk models and cutting latency from 4 hours to under 20 minutes.

EXPERIENCE

CVS, USA | Data Engineer,

Jan 2025 - Current

- Built scalable PySpark pipelines on **Databricks** to clean and process 120M+ pharmacy and delivery records weekly, powering stockout risk models and improving restock planning accuracy by 22%.
- Designed modular SQL models in **Azure Synapse** to track refill lag, overstock trends, and supplier penalties; helped cut average delivery delays by 1.4 days across 3 distribution zones.
- Automated 9 high-usage **Power BI dashboards** with row-level security and real-time refreshes, giving warehouse ops and regional finance teams visibility into order flow, cost per shipment, and fill rate trends.
- Used **Python (pandas, matplotlib)** to run deep-dive analysis on cold chain failures and expiration-driven write-offs, surfacing \$1.1M in avoidable losses over the past 6 months.
- Tuned Spark job configs and switched to **Kryo serialization**, reducing ETL runtime by 29% and bringing down monthly Databricks compute spend by \$9.2K.

Kensoft Inc | Data Engineer

June 2022 - May 2023

- Engineered high-volume **PySpark pipelines** on Databricks to process 70M+ daily records across trades, payments, and holdings; brought pipeline run-time down from 3.5 hours to under 90 minutes.
- Built normalized and star-schema data models in **PostgreSQL**, supporting accurate joins across 12+ tables spanning instruments, accounts, and counterparty mappings.
- Scheduled and monitored 20+ critical jobs in **Airflow**, including dependency chains, SLA alerts, and failure reruns; improved batch reliability to 99.2% over 6 months.
- Tuned long-running SQL queries using **partitioning, indexing, and materialized views**, cutting report load times from 12 minutes to under 2 minutes for treasury and reconciliation teams.
- Integrated S3-based FX feeds with staging tables using **AWS Lambda**, ensuring daily currency conversion logic ran under 5 minutes across all risk reporting pipelines.
- Designed secure data marts in **Snowflake** using dbt for position snapshots and margin exposures, enforcing role-based access for regulatory audits and internal controls.
- Migrated 40+ legacy Excel-based reports into structured **Power BI dashboards**, reducing manual prep by 80% and giving real-time insights into margin calls and daily PnL swings.

TECHNICAL SKILLS

- **Programming Language:** Python(NumPy, Pandas, SciPy, PySpark, Scikit-learn), R, SQL, SAS, Macros
- **Data Engineering & Cloud Technologies:** AWS, Azure Data Factory, Kubernetes, REST APIs, Model Serving, ETL, Data Warehousing, Data Lake, Hadoop, Spark Streaming, Kafka, Alteryx, Glue, Hive, Athena, MapReduce,
- **Data Visualization & Business Intelligence:** Data Reporting, Dashboard Development, Data Storytelling, Data Interpretation, Trend Analysis, Cohort Analysis, A/B Testing, Customer Segmentation, Geospatial Analysis, Data Mining
- **Database Management:** PostgreSQL, MySQL, NoSQL, Data Modeling, Database Design, Data Migration, Data Integration, Data Governance, Data Security, Data Encryption
- **Model Deployment & Monitoring:** CI/CD, Jenkins, Terraform, Model Retraining, Performance Tracking, Anomaly Detection, Alerting, Version Control, Continuous Integration, Continuous Delivery

EDUCATION

Kent State University

Master of Science in Computer Science

Kent, OH

May – 2025